

**Haldia Institute of Technology**  
**Dept: Computer Science and Engineering**  
**2<sup>nd</sup> Class Test – May 2026**

**Branch: CSE**

**Year: 2nd**

**Batch 1+2+3**

**Semester: 4th**

**Paper Name: Object Oriented Programming**

**Paper Code: PCC-CS 402**

**Full Marks: 15**

**Time: 50 Min**

- 
- 1 ✓ Create a superclass called Shape with the following methods: [CO-2] 5

getxyvalue() to accept the coordinates (length and breadth).

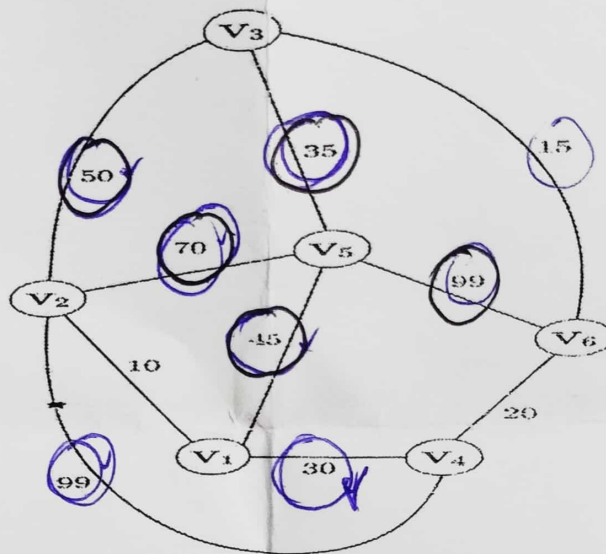
showxyvalue() to display the coordinates.

Create a subclass called Rectangle that inherits from Shape.

Override the showxyvalu() method in the Rectangle class to display the length and breadth of the rectangle.

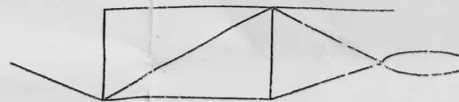
- 2 ✓ Explain the use of try, catch, and finally blocks with syntax. [CO-3] 5
- 3 ✓ Write sample Java program to implement Arithmetic Exception. [CO-3] 5

1. Use Prim's algorithm to find a minimal spanning tree from the graph given below: (CO 2)



2. Prove that the number of pendent vertices in a binary tree with  $n$  vertices is  $\frac{(n+1)}{2}$ .  
(CO 2)

3. Draw the dual of the graph: (CO 3)



--END--

**Haldia Institute of Technology**  
**Department of Computer Science & Engineering**  
**EVEN Sem. Class Test- II, May 2026**

Year: 2nd

Semester: 4th

Paper Name: Machine Learning

Paper Code: PCC-CSE 403

Full Marks: 15

Time:

Answer the following questions:

[3 x 5]

1. When does entropy become zero? Explain terms Information Gain and Gini Index with formula. (2+3) [CO:3]

OR

What is F1-score? Why is it useful in evaluating classification models?

2. Why is conditional independence important in Naive Bayes? Discuss the advantages and disadvantages of Naive Bayes algorithm. (2+3) [CO:3]

OR

Write the differences between linear and non-linear SVM? Explain the strengths and weaknesses of the SVM algorithm.

3. Evaluate the information gain for features **CGPA** and **Communication** from the below dataset. (5) [CO:2]

| CGPA   | Programming Skill | Communication | Job Offered? |
|--------|-------------------|---------------|--------------|
| High   | Good              | Good          | Yes          |
| Medium | Good              | Good          | Yes          |
| High   | Good              | Bad           | Yes          |
| High   | Good              | Good          | Yes          |
| High   | Bad               | Good          | Yes          |
| Medium | Good              | Good          | Yes          |
| Low    | Bad               | Bad           | No           |
| Low    | Bad               | Bad           | No           |
| Medium | Good              | Bad           | Yes          |
| Medium | Bad               | Good          | No           |
| Medium | Good              | Bad           | Yes          |

OR

Apply k-means algorithm in given data for k=2 (that is 2 clusters).

Use C1(80) and C2(250) as initial cluster centers.

Data: 234, 123, 456, 23, 34, 56, 78, 90, 150, 116, 117, 118, 199.

OR

A binary classification model produces the following confusion matrix values: TP = 35, FP = 15, TN = 45, FN = 5. State the terms **Precision** and **Recall** and calculate the values of these metrics for the model using appropriate formulas.

Haldia Institute of Technology  
Department of Computer Science & Engineering  
EVEN Sem. Class Test- II, May 2026

Year: 2nd  
Paper Name: Database Management Systems

Semester: 4<sup>th</sup>, Batch: I, II, III  
Paper Code: PCC-CSE 401

Full Marks: 15

Time: 50 Min.

Answer the following questions:

1. ✓ Explain projection operation in relational algebra with suitable example.

[CO2]

or

OR

✓ Explain Functional dependency with suitable example.

2. ✓ Explain the pros and cons of Sequential File Organization.

[CO3]

✓ 3. Explain the concept of Hash File Organization with suitable diagram.

[CO3]



**Haldia Institute of Technology**  
**Department of Computer Science & Engineering**  
**4<sup>th</sup> Sem. Class Test-II, May 2026**

**Subject:** Formal Language & Automata Theory

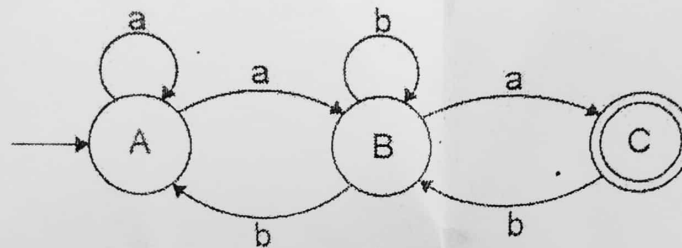
**Subject Code:** PCC-CSE 405

**Full Marks:15**

Time: 50 min.  
[3x5]

**Answer all three questions:**

- ✓ 1. Obtain the regular expression that denotes the language accepted by the following DFA using Arden's Theorems. [CO2]

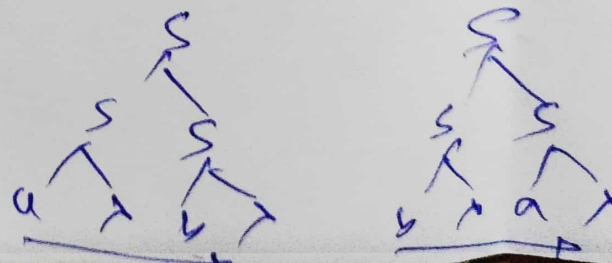


2. Consider the grammar-  
 $S \rightarrow bB/aA$

$$A \rightarrow b/bS/aAA$$
$$B \rightarrow a/aS/bBB$$

For the string  $w = \text{bbaababa}$ , find the **Left Most Derivation**, **Right Most Derivation** and **Parse Tree**. [CO3]

3. Check whether the given grammar is ambiguous or not-  
 $S \rightarrow SS$

$$S \rightarrow a$$
$$S \rightarrow b$$


[C03]



**Haldia Institute of Technology**  
**(An Autonomous Institute)**  
**2<sup>nd</sup> Class Test – May, 2026**

**Programme: CSE,**

**Semester: 4<sup>th</sup>**

**Paper Name: Industrial Management**  
**Paper Code: HSMC-CSE 401**

**Full Marks: 15**

---

**Attempt all the questions.**

**5X3 = 15 Marks**

✓ 1. What the barriers to effective planning?

**[CO-3]**

✓ 2. What are the various steps to effective decision making?

**[CO-3]**

✓ 3. What are the advantages and disadvantages of Decentralization?

**[CO-4]**